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Monocular Human Mesh Reconstruction from Sparse 3D Keypoints



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Introduction

1.1 Motivation

Human mesh recovery (HMR) aims to reconstruct the 3D surface geometry of the human body from images or videos. Applications of HMR include autonomous driving, security and surveillance or virtual and augmented reality [1]. Compared with pose estimation, which represents the human body using sparse 2D or 3D joints, HMR reconstructs the body surface as a 3D mesh and therefore provides a finer-grained representation of human shape and pose, which is commonly achieved using a parametric human body model [2]. Modern HMR approaches frequently formulate the problem as a regression task, using deep neural networks to infer body model parameters or mesh representations from RGB image features [3]. However, recovering a 3D human mesh from RGB images is challenging because of the domain gap between 2D image data and 3D geometry [2].

To address this challenge, several methods, such as HybrIK [4], SMPLR [5], and I2L-MeshNet [6], first estimate sparse 3D keypoints as an intermediate representation before reconstructing the final human mesh. This intermediate representation separates image understanding from mesh reconstruction, which avoids the need to directly predict abstract mesh parameters from RGB images.

However, existing methods typically combine sparse 3D keypoints with additional visual or geometric cues. For example, HybrIK [4] and I2L-MeshNet [6] exploit image features in addition to 3D keypoints during mesh estimation, while SMPLR [5] augments sparse joint coordinates with additionally predicted surface landmarks. Therefore, it remains unclear to what extent sparse monocular 3D keypoints alone retain sufficient information to recover both human pose and body shape.

This thesis investigates this question by reconstructing SMPL [7] pose and shape parameters solely from 3D keypoint sequences predicted by MotionBERT [8], without additional input information.

1.2 Research Questions

1. **How well can SMPL pose parameters be reconstructed from sparse 3D keypoints using a learned regression network?**

   To answer this research question, a straightforward learning-based regressor is trained and its performance is compared against an optimization-based baseline.
2. **Can SMPL body shape parameters be reconstructed from sparse 3D keypoint sequences using a temporal network exploiting motion dynamics?**

   This question investigates whether body shape can be inferred from temporal patterns in the 3D joint motion. A model exploiting motion dynamics is trained and compared against a baseline that uses a neutral shape.
3. **How does the performance of a human mesh reconstruction pipeline based on sparse 3D keypoints compare to a direct video-to-mesh baseline (VIBE)?**

   This question compares the proposed pipeline using sparse 3D pose coordinates as an intermediate representation against a more traditional direct video-to-mesh approach represented by VIBE [9]. The comparison focuses on differences in pose accuracy and temporal smoothness.

1.3 Outline

The remainder of this thesis is structured as follows: Section **2** describes the dataset, the theoretical components of the proposed pipeline, and evaluation details. Section **3** covers implementation details, including 2D keypoint extraction, 3D pose lifting, the setup of the optimization-based baseline for pose reconstruction, as well as the architectures of the pose and shape parameter estimators. Section **4** presents the experimental results addressing the research questions, Section **5** discusses an alternative approach for HMR using sparse 3D keypoints, and Section **6** concludes the thesis with a summary of the findings, possible improvements for the proposed pipeline, as well as an outlook on future research.

Material and Methods

2.1 Material

2.1.1 3D Poses in the Wild Dataset (3DPW)

3DPW [10] is a dataset for 3D human pose estimation and human mesh reconstruction. It includes 60 video sequences (over 51,000 frames) captured using a handheld camera, depicting people performing everyday activities in a variety of indoor and outdoor environments. The dataset includes seven distinct actors, with clothing varying across sequences. It provides annotations such as 2D joint coordinates, camera parameters, and SMPL [7] shape and pose parameters. The SMPL model is further described in Section 2.2.1. For 3DPW, body shape was captured once per actor using 3D scans. Pose was reconstructed by combining IMU signals, 2D keypoints, and estimated camera poses (camera position and orientation over time). The included sequences have up to two annotated subjects per sequence and frequently feature background subjects.

3DPW is widely used as a benchmark dataset for evaluating 3D human pose estimation methods [1].



Figure 2.1: 3DPW samples (cropped) with ground-truth SMPL [7] overlay.

2.2 Methods

2.2.1 Skinned Multi-Person Linear Model (SMPL)

The Skinned Multi-Person Linear Model (SMPL) [7] is a parametrized representation of the human body, learned from a large dataset of 3D human scans. It defines a human body as a function of pose $\theta \in \mathbb{R}^{72}$ and shape $\beta \in \mathbb{R}^{10}$. The pose parameters represent the relative rotations of 23 joints and the global rotation of the pelvis (root joint) in axis-angle representation, while the shape parameters correspond to 10 principal components of body shape.
We use the *official code implementation*[1] provided by the authors, where the parametric model is initialized by a forward pass given the pose and shape parameters.

2.2.2 AlphaPose

AlphaPose [11] is a top-down, multi-person pose estimation framework. Given an image input, a human detector first predicts bounding boxes for each person, which are then processed individually by a Single-Person Pose Estimator (SPPE). FastPose is the SPPE within the AlphaPose pipeline. The SPPE uses a ResNet [12] backbone followed by three Dense Upsampling Convolution (DUC) [13] layers to upsample the extracted feature maps before a final convolutional layer generates keypoint heatmaps. Instead of conventional heatmap peak detection, the AlphaPose framework introduces Symmetric Integral Keypoint Regression (SIKR), which predicts keypoint coordinates as the weighted average of the corresponding heatmap. To remove duplicate pose estimates caused by overlapping detections, the authors propose pose-based Non-Maximum Suppression (Pose NMS), which filters redundant poses using a pose-level similarity metric. The authors demonstrate that their framework provides a strong trade-off between inference speed and pose estimation accuracy.

AlphaPose is recommended in the *official code implementation of MotionBERT*[2] for 2D keypoint extraction, as it delivers keypoint predictions in the Halpe 26 format, which is the native input format of MotionBERT.

For this project, we use AlphaPose with the FastPose *Halpe 26 keypoints SPPE*, which was trained on the Halpe dataset introduced in the AlphaPose paper [11]. The model predicts 26 body joints, visualized in Figure 2.2 (left).

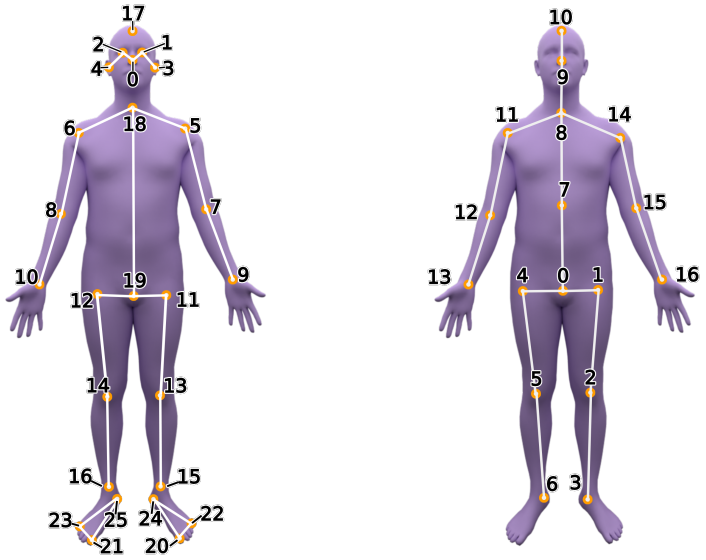
2.2.3 MotionBERT

MotionBERT [8] is a transformer-based 3D pose estimation framework. It does not operate directly on pixels but instead requires a 2D pose estimator as a backbone. Natively, it takes 2D-keypoint sequences in the Halpe 26 format as input. The extracted keypoint trajectories are processed by the dual-stream spatio-temporal transformer (DSTformer) architecture introduced by the authors,

[1] Official SMPL repository: https://github.com/vchoutas/smplx
[2] Official MotionBERT repository: https://github.com/Walter0807/MotionBERT

which models relationships between body joints within each frame (spatial attention) and across time (temporal attention). Through this spatio-temporal modeling, the DSTformer learns a motion embedding, on which a linear transformation is then applied to infer the final 3D joint coordinates. At the time of its publication, MotionBERT achieved state-of-the-art performance on 3D human pose estimation benchmarks. For each frame, MotionBERT predicts 17 3D joint coordinates in the Human3.6M (H3.6M) [14] keypoint format, visualized in Figure 2.2 (right).

Although more recent methods have reported slightly better performance, MotionBERT remains among the top-performing approaches [15].



Halpe 26 Format H3.6M Format (2D-projected)

Figure 2.2: Keypoint formats used in the pose estimation pipeline overlaid onto the SMPL [7] 3D model (own illustration).

2.2.4 VIBE

VIBE (Video Inference for Human Body Pose and Shape Estimation) [9] is a deep learning-based approach for 3D human pose and shape estimation from monocular RGB images or videos.
Although more recent methods have surpassed its performance, VIBE remains a widely used baseline and reference model in subsequent video-based mesh recovery work, including TCMR [16] and WHAM [17].

The framework proposed by Kocabas et al. [9] estimates SMPL parameters directly from an RGB video. The pipeline first extracts image features using a pretrained Convolutional Neural Network (CNN). A Gated Recurrent Unit (GRU) [18] network then combines information from several consecutive frames, such that the model can use temporal information. After that, VIBE uses a regressor with fully connected layers to predict the SMPL body and camera parameters. Camera parameters are used to project the predicted mesh onto

the image plane during training to calculate the loss. For this purpose, VIBE uses a weak-perspective camera model, a simplified camera model that represents the projection from 3D to 2D mainly by scaling and shifting the body. In addition, the temporal network and the motion-based training help the model produce smoother and more realistic human motion.

In this work, VIBE serves as a representative reference model for comparing the proposed 3D keypoint-based method with an approach that directly infers SMPL parameters from video.

2.2.5 Evaluation

Evaluation Metrics

Although the evaluation metrics used in this work are common in the literature, their definitions and implementations can vary across papers, e.g. in alignment, joint selection, or the handling of scale and translation [19]. To ensure consistency, all metric definitions below follow the VIBE evaluation protocol described in the original paper [9] and implemented in the corresponding *official code implementation*[3].
Following this protocol, coordinates are assumed to be in meters, with errors converted to millimeters by a factor of 1000. The SMPL model is initialized with the predicted pose and shape parameters without translation, making the evaluation translation-independent. The VIBE mesh-to-joints regressor `J_regressor_h36m` is applied to the predicted and ground-truth meshes to obtain 17 joints, of which 14 are evaluated. Introduced in SMPLify [20], this subset of 14 keypoints provides a common basis across different joint representations and allows for cross-method comparison.

Mean Per Joint Position Error (MPJPE)

MPJPE measures the mean Euclidean distance between the predicted and ground-truth 3D joint positions, $\tilde{\mathbf{p}}_{t,j}$ and $\hat{\mathbf{p}}_{t,j}$, obtained by applying the joint regressor to the predicted and ground-truth meshes. Both sets of joint positions are root-aligned by subtracting their respective pelvis positions $\tilde{\mathbf{c}}_t$ and $\hat{\mathbf{c}}_t$. The resulting error is averaged over all evaluated frames and joints. For T evaluated frames and N_j joints ($N_j = 14$ for VIBE on 3DPW), the metric is mathematically formalized below based on the *official code implementation*[3], matching the evaluation protocol described in the original paper [9]:

$$\mathrm{MPJPE}_{\mathrm{mm}} = \frac{1000}{TN_j} \sum_{t=1}^{T} \sum_{j=1}^{N_j} \left\| (\tilde{\mathbf{p}}_{t,j} - \tilde{\mathbf{c}}_t) - (\hat{\mathbf{p}}_{t,j} - \hat{\mathbf{c}}_t) \right\|_2 , \tag{2.1}$$

Procrustes-Aligned MPJPE (PA-MPJPE)

PA-MPJPE is computed similarly to MPJPE, but after additionally applying a Procrustes [21] alignment to the pelvis-centered predicted joints $(\tilde{\mathbf{p}}_{t,j} - \tilde{\mathbf{c}}_t)$. This

[3]Official VIBE repository: https://github.com/mkocabas/VIBE

yields aligned predicted joints $\tilde{\mathbf{p}}_{t,j}^{\text{aligned}}$, which are then used in place of the pelvis-centered predicted joints when computing MPJPE. This aligns the predicted set of joints with the ground-truth set of joints in terms of scale s, rotation $\mathbf{R}$, and translation $\mathbf{d}$. For each frame t, we formalize the optimal similarity transformation according to the *official code implementation*[3], which follows the evaluation procedure described by the authors of VIBE [9]:

$$(s_t^*, \mathbf{R}_t^*, \mathbf{d}_t^*) = \arg\min_{s_t, \mathbf{R}_t, \mathbf{d}_t} \sum_{j=1}^{N_j} \left\| s_t \mathbf{R}_t (\tilde{\mathbf{p}}_{t,j} - \tilde{\mathbf{c}}_t) + \mathbf{d}_t - (\hat{\mathbf{p}}_{t,j} - \hat{\mathbf{c}}_t) \right\|_2^2, \tag{2.2}$$

$$\tilde{\mathbf{p}}_{t,j}^{\text{aligned}} = s_t^* \mathbf{R}_t^* (\tilde{\mathbf{p}}_{t,j} - \tilde{\mathbf{c}}_t) + \mathbf{d}_t^*. \tag{2.3}$$

Per-Vertex Error (PVE)

The PVE, also called Mean Per Vertex Error (MPVE), measures the average Euclidean distance between the estimated 3D mesh vertices $\tilde{\mathbf{x}}$, and their ground-truth counterparts $\hat{\mathbf{x}}$. VIBE predicts a shape parameter vector for each frame, even if it is expected to be temporally consistent for one person across frames [9]. In contrast, the method proposed in this work estimates a single shape vector per sequence. To maintain consistency with VIBE, we compute PVE over all T frames and all V mesh vertices ($V = 6890$ for the SMPL body model). The PVE formulation is derived from the *official code implementation*[3], following Kocabas et al. [9]:

$$\text{PVE}_{\text{mm}} = \frac{1000}{TV} \sum_{t=1}^{T} \sum_{v=1}^{V} \left\| \tilde{\mathbf{x}}_{t,v} - \hat{\mathbf{x}}_{t,v} \right\|_2 \tag{2.4}$$

Evaluation Procedure

The evaluation follows the VIBE protocol and is performed in camera coordinates. Consequently, the SMPL root orientation is expected in the camera frame. Since 3DPW provides root orientations in the global frame, the official VIBE preprocessing converts them to camera-relative rotations [9].
In our pipeline, the root orientation is estimated from MotionBERT keypoints. MotionBERT predicts these keypoints in a normalized coordinate system based on the initial bounding-box crop rather than the camera frame [8]. Consequently, the inferred root rotations are expressed in this normalized space, which may affect rotation-sensitive metrics such as MPJPE and PVE (Section 2.2.5).
This normalization does not remove body orientation. The predicted MotionBERT skeletons still vary with the subject's orientation relative to the camera rather than being mapped to a canonical body orientation [8]. We therefore report the rotation-sensitive metrics while noting that they may reflect both pose errors and the mismatch between normalized and camera coordinate systems.
Evaluation uses the VIBE-defined test split of 34,561 person-frames, counting each annotated subject separately, as derived from the *VIBE evaluation and preprocessing code* [3].

Implementation

3.1 2D Keypoint Extraction

We employ AlphaPose [11] in its default configuration for 2D keypoint extraction. Since it predicts poses for all detected people, the dataset's target subject must be identified before passing the keypoints to MotionBERT [8], which expects a single-person keypoint sequence as input.
For this step, we reuse the default bounding box preparation pipeline from the *VIBE evaluation code*[3]: ground-truth 2D joints define per-frame bounding boxes around the target subject, missing detections are linearly interpolated, the box parameters are temporally smoothed, and the final regions are then converted into square image crops.
AlphaPose then resizes each crop, normalizes the pixel values to produce a 256×192 input tensor and processes it with its SPPE. It refines the predictions as described in Section 2.2.2 to obtain the final 26 2D coordinates in the original image coordinate system [11].

3.2 3D Pose Prediction

For 3D pose prediction, we use the pre-trained MotionBERT [8] model in its default form. Before lifting the keypoints, MotionBERT's built-in format mapping converts the AlphaPose 2D keypoints from Halpe-26 to Human3.6M, handling missing correspondences through exclusion or approximation from adjacent joints. MotionBERT normalizes the keypoints to $[-1, 1]$ using a square region whose side length is determined by the larger pose dimension [8].
Consequently, image position and scale are removed, so the resulting 3D joint positions are expressed in a normalized coordinate space.
To capture temporal context, the normalized trajectories are divided into contiguous clips using MotionBERT's default `clip_len = 243` and processed by its spatio-temporal transformer to predict the 3D joint positions [8].

Our own implementation for 2D keypoint extraction and 3D pose prediction is limited to managing the data flow between AlphaPose and MotionBERT and automating execution across the full dataset.

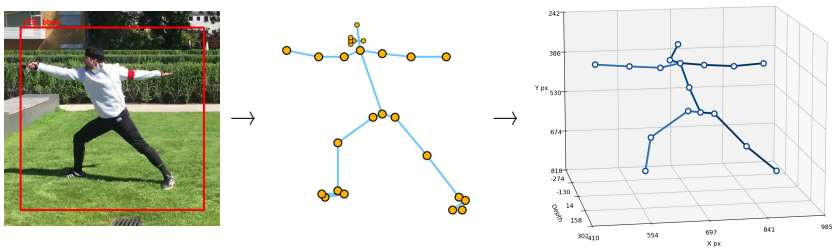


Figure 3.1: 2D-to-3D keypoint workflow: bounding box selection, AlphaPose 2D keypoints, and MotionBERT 3D joints.

3.3 Pose Parameter Estimation

To infer the final SMPL parameters from the predicted 3D keypoints, we primarily investigate a regression network. For comparison, we additionally employ a simple optimization-based baseline. Their performance is compared in the experimental results (Section 4.1). Since this section focuses on pose estimation, neutral shape parameters ($\beta = 0$) are assumed throughout.

3.3.1 Method 1: Optimization (Baseline)

One approach for estimating SMPL parameters from an intermediate representation is the optimization-based method SMPLify [20], which minimizes the reprojection error between detected 2D keypoints and projected 3D SMPL joints while enforcing prior-based regularization.

Inspired by SMPLify at the conceptual level, we formulate SMPL parameter estimation as an optimization problem, without reusing SMPLify's objective function, optimizer, or released implementation. While SMPLify minimizes a 2D reprojection error, we define a simple 3D joint-alignment objective using only the unweighted mean-squared-error component of the joint error term described in the supplementary material of Kaufmann et al. [22]. The resulting objective minimizes the mean squared distance between root-centered MotionBERT keypoints and the corresponding root-centered SMPL joints. The SMPL joints are obtained using the evaluation joint regressor described in Section 2.2.5. We implement the optimization in PyTorch [23] using automatic differentiation and the Adam optimizer [24]. For each frame t, we optimize the pose parameters θ_t^* while fixing the shape parameters to the neutral shape, $\beta = \mathbf{0}$. The optimization problem can therefore be described as [22]:

$$\theta_t^* = \underset{\theta}{\operatorname{argmin}} \frac{1}{J} \sum_{j=1}^{J} \left\| \tilde{X}_{t,j}^{\mathrm{SMPL}} - \hat{X}_{t,j}^{\mathrm{MB}} \right\|_2^2, \tag{3.1}$$

where the model's 3D joints $X_{t,j}^{\mathrm{SMPL}}$ are obtained by applying the H3.6M joint regressor W^{H36M} to the generated SMPL body mesh, following the notation of SPIN [25]:

$$X_t^{\mathrm{SMPL}} = W^{\mathrm{H36M}} M(\theta_t, \beta_0). \tag{3.2}$$

Here, t and j index the frames and the 17 H3.6M joints, respectively. θ_t^* denotes the optimized pose parameters for frame t, while β_0 represents the fixed, unoptimized default shape.

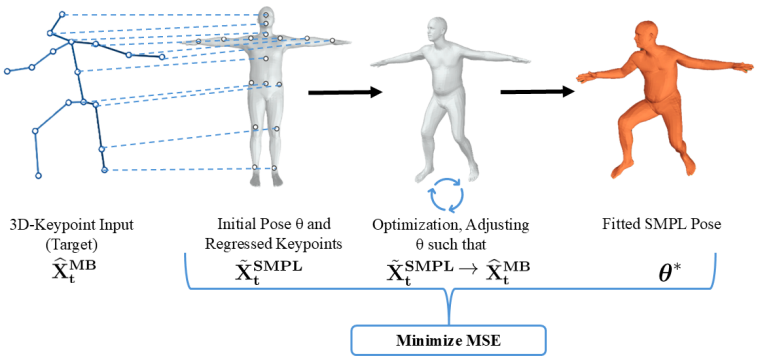


Figure 3.2: Workflow of the optimization-based SMPL parameter estimation. Own illustration using the MotionBERT [8] output and the SMPL [7] 3D model.

3.3.2 Method 2: Deep Learning

As our second, and primary, approach, we employ the pose branch of the SMPL reverse stage proposed by the authors of SMPLR [5]. The pose branch of the original network is a multilayer perceptron (MLP) with batch normalization and ReLU activations in each hidden layer. Its input consists of root-centered bone vectors and predicted mesh landmarks. After five hidden layers, following a dual-head refinement design, the network generates a main prediction and an offset from an intermediate layer, which are fused through concatenation, a final fully connected layer, and a tanh activation. The network outputs 24×9 raw rotation matrices, representing the 24 joint poses as 3×3 rotation matrices rather than axis-angle vectors. The authors modify the SMPL model to accept rotation matrices directly during the forward pass. Training utilizes three L_1 loss terms to supervise the predicted joint rotations ($\mathcal{L}_R$), SMPL-regressed joints ($\mathcal{L}_{joints}$), and predefined mesh vertex landmarks ($\mathcal{L}_{landmarks}$), respectively. The combination of the latter two is referred to as $\mathcal{L}_{SMPL}$ [5]:

$$\mathcal{L} = \mathcal{L}_R + \mathcal{L}_{joints} + \mathcal{L}_{landmarks}. \tag{3.3}$$

We adapt the original model in three ways. First, we replace the input layer with one that accepts root-centered bone direction vectors (16×3) computed from the MotionBERT keypoints, instead of the original 24×3 bone vectors. Second, we omit the predicted mesh landmarks from the network input. Finally, since we consider only pose estimation, we fix the SMPL shape parameters to the neutral body shape on both the prediction and ground-truth sides ($\beta = \mathbf{0}$). In contrast, the original work uses a separate model branch to output shape parameters. The adapted network architecture used is illustrated in Figure 3.3.

Following the VIBE training/validation split, the dataset comprises 22,663 training and 10,261 validation samples, as can be inferred from the corresponding *code implementation* [3]. We implement the SMPLR-inspired network architecture and training procedure ourselves in PyTorch, including our own choice of training hyperparameters, while reusing the *official SMPLR code implementation*[4] for their modified version of the SMPL model and the training loss defined in Equation 3.3. The network was trained with the Adam optimizer for 40 epochs using a batch size of 256 and a learning rate of 10^{-3}. As seen in Figure 3.3, the validation loss decreases sharply during the first epochs before stagnating after around 10 epochs, indicating that the model learns the main pose mapping quickly and that later training delivers only limited generalization gains. As the training loss continues to decrease, the gap between the training and validation loss widens, which suggests that the model is overfitting.

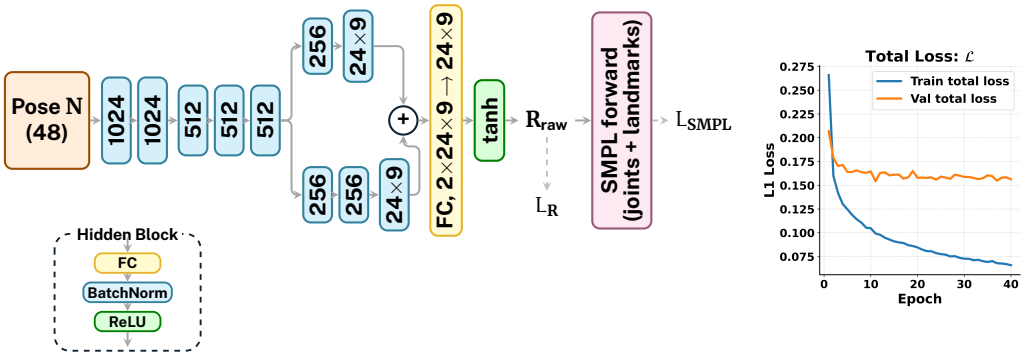


Figure 3.3: Deep learning-based SMPL pose estimator. Left: Adapted pose branch of the SMPL reverse stage from SMPLR [5]. Right: Total training and validation loss.

3.4 Shape Parameter Estimation

Methods that regress body shape from 2D or 3D keypoints commonly use anthropomorphic cues such as bone length ratios. For example, SMPLR incorporates bone lengths as a constraint for shape estimation [5].
However, MotionBERT is trained to estimate accurate joint positions for pose reconstruction rather than to preserve shape-related differences between subjects. Therefore, bone length ratios derived from its predictions may not provide sufficient shape cues. Instead, we investigate whether body shape can be inferred from motion dynamics of predicted 3D keypoints.
Previous work in biometrics has shown that gait and body motion encode traits such as gender, age, and body type [26]. Motivated by this, we estimate SMPL shape parameters from temporal sequences of MotionBERT keypoints.

[4]Official SMPLR repository: https://github.com/meysam-madadi/SMPLR

To model these temporal dynamics, we adopt the core design principles of the Residual Temporal Convolutional Network (Res-TCN) architecture proposed by Kim et al. [27]. The original network consists of an initial convolutional block followed by residual units for temporal feature extraction. According to the authors, the initial convolutional block extracts local temporal features, which are progressively refined by the residual units.

We adapt the architecture to our regression task by modifying the convolutional filter sizes, channel dimensions, and the number and configuration of the residual units. Furthermore, unlike the original classification network, we replace the final softmax classifier with a regression head consisting of two fully connected layers with ReLU activation and dropout. As shown in Figure 3.4, the resulting network takes raw, non-root-centered 3D keypoint sequences as input and contains an initial convolutional block, eight residual blocks, global average pooling, and the regression head, which predicts the 10 SMPL shape parameters from a 256-dimensional feature vector. The network is trained using an L_1 loss between the predicted and ground-truth shape parameters β.

The train/validation split is identical to that used for the pose model (Section 3.3.2). We implement the adapted network in PyTorch and select its architectural and training hyperparameters ourselves.

We use overlapping clips of $T = 50$ frames with a stride of $s = 25$, train the network for 80 epochs with the Adam optimizer, set the batch size to 64 and the learning rate to 10^{-3}. Figure 3.4 shows the training curves. While the training loss decreases, the validation loss plateaus and exhibits noticeable fluctuations, suggesting limited generalization to unseen subjects. The gap between training and validation loss indicates overfitting.

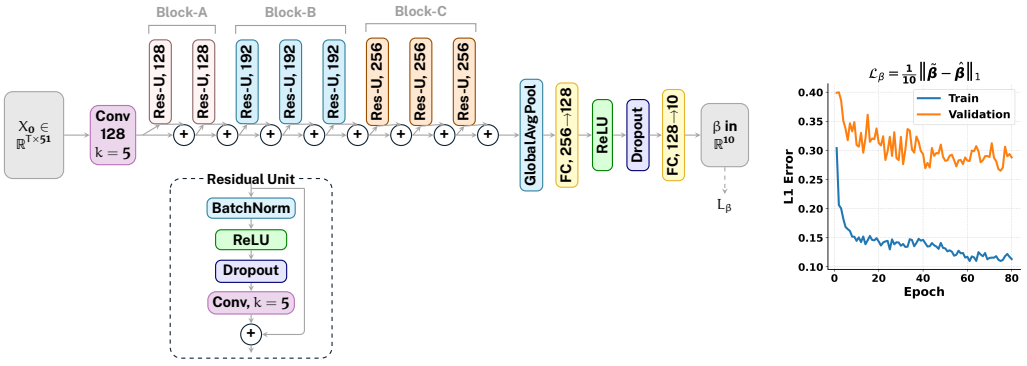


Figure 3.4: Temporal SMPL [7] shape estimator. Left: architecture of the temporal TCN adapted from Kim et al. [27] used for shape regression. Right: training and validation loss curves.

Experimental Results

4.1 SMPL Pose Parameter Reconstruction (RQ 1)

How well can SMPL pose parameters be reconstructed from sparse 3D keypoints using a learned regression network?

To isolate the pose component of SMPL, the shape parameters β are set to zero for all methods on both the prediction and ground-truth sides during evaluation in this research question.

Table 4.1: Comparison of 3D-to-SMPL pose estimation accuracy (PA-MPJPE, MPJPE, PVE in mm) evaluated under a neutral-shape setting ($\beta = \mathbf{0}$).

	PA-MPJPE ↓	MPJPE ↓	PVE ↓
VIBE (SOTA baseline)	50.43	75.47	89.21
Optimization	80.12	132.44	164.26
Adapted SMPLR	62.11	95.54	111.27

Optimization: The results of the baseline optimization run, described as Method 1 (Optimization) in Section 3.3.1, suggest that, as expected, the objective function is underconstrained for this task. The reconstructed poses show unnatural body configurations and implausible bending in most frames, as shown in Figure 4.1. Since minimizing the MSE between corresponding joint locations does not uniquely determine the underlying SMPL parameters, the optimization can converge to anatomically incorrect poses. Specifically, axial rotations of body parts are not constrained, which can cause visually unrealistic solutions with similarly low objective values.

Deep Learning (Adapted SMPLR): The deep learning approach, which corresponds to the adapted SMPLR model described as Method 2 (Deep Learning) in Section 3.3.2 (labeled as *Adapted SMPLR* in Table 4.1), outperforms the optimization baseline across all measured metrics. The adapted SMPLR architecture works well for the problem of regressing pose parameters, despite its rather simple architecture.
The regression problem is simplified by operating on the 3D skeletal representation rather than directly on image features, as many other approaches do. Since the input already contains the 3D skeletal structure, the mapping to local joint rotations becomes a less complex regression task. This indicates that a simple multilayer perceptron is sufficient to plausibly estimate the SMPL pose parameters. However, due to the sparse input, finer pose details, such as wrist or ankle rotation cannot always be recovered reliably.

This experiment shows that even with a simple MLP, we were able to infer plausible pose parameters based on the sparse 3D coordinate input.

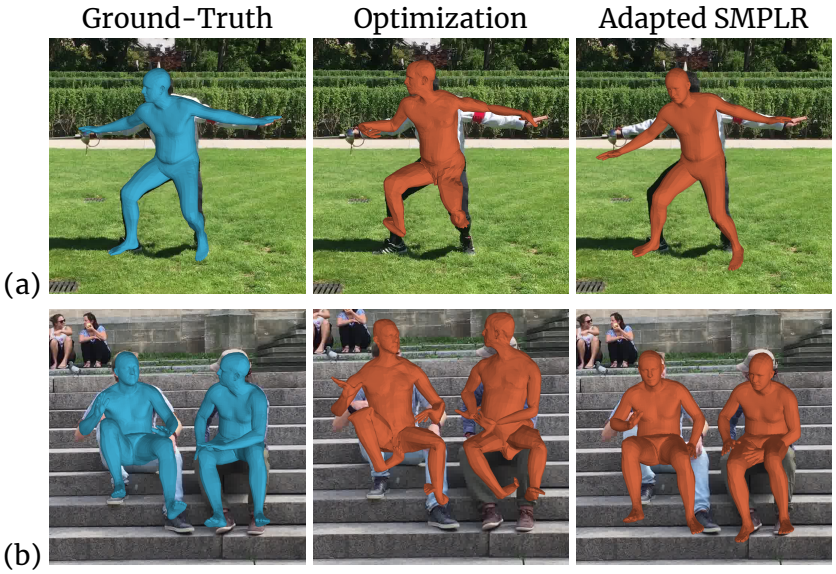


Figure 4.1: Qualitative comparison of 3D-coordinate-to-SMPL pose-estimation methods for two sample frames (neutral shape for both ground-truth and predictions).

4.2 SMPL Shape Parameter Reconstruction (RQ 2)

Can SMPL body shape parameters be reconstructed from sparse 3D keypoint sequences using a temporal network exploiting motion dynamics?

To obtain the results of this research question, evaluation is performed with pose parameters θ set to zero for all methods on both prediction and ground-truth sides to eliminate the influence of pose on the results.
For shape-only evaluation, we report only PVE, as joint errors such as MPJPE are not informative for evaluating shape accuracy. Vertex errors are better suited for this evaluation [28].

As for the previous research question, we include VIBE as a reference. Although VIBE predicts shape parameters for each frame, body shape is assumed to remain constant throughout a sequence, and the ground-truth provides only a single shape vector per sequence. Therefore, for VIBE we use the median of the predicted shape parameters across all frames to obtain a single shape estimate matching our model output.

Table 4.2: Comparison of body shape estimation accuracy (PVE in mm) evaluated under a neutral-pose setting ($\theta = \mathbf{0}$).

Method	PVE ↓
VIBE (SOTA baseline)	32.85
Baseline ($\beta = \mathbf{0}$)	28.78
Temporal TCN	26.84

In this evaluation, *Temporal TCN* refers to the temporal SMPL shape estimator implemented in Section 3.4, while the *Baseline* ($\beta = 0$) uses the default neutral SMPL body shape with $\beta = \mathbf{0}$. As shown in Table 4.2, the proposed method

performs slightly better than the neutral-shape baseline. However, the differences between the resulting metrics are relatively small and therefore provide only limited insight into the quality of the predicted shape parameters. To further assess the behavior of the shape estimator, we evaluate the consistency of its predictions across multiple test sequences belonging to the same actor, as shown in Figure 4.2. One actor appears in 20 test sequences and another in 14, allowing us to examine whether the model produces similar shape estimates for the same individual across different sequences.

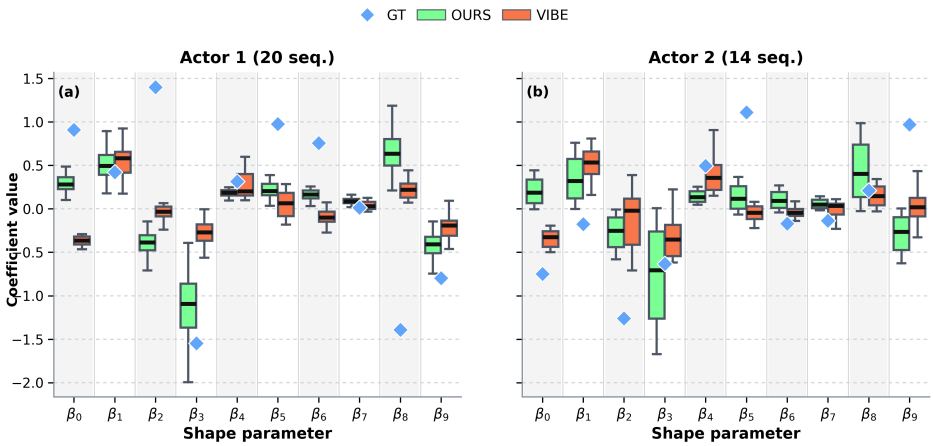


Figure 4.2: Analysis of shape prediction consistency for the top two actors in the 3DPW test set. Narrower spreads indicate more consistent shape predictions.

Figure 4.2 shows that our method exhibits inconsistent predictions for several shape parameters, such as β_3 for both actors and β_8 for actor 2. Furthermore, the estimates appear to be only weakly influenced by the ground-truth, as the median of most predicted parameters remains at a similar value for both actors. This suggests that the model captures subject-specific body shape only to a limited extent. A likely explanation is that motion dynamics provide insufficient shape cues, while the model architecture and training setup, including the small kernel size of 5, may not capture the subtle relationship between motion and body shape. Consequently, the predicted shape parameters tend to converge to similar values across subjects, regardless of the true underlying shape.

A constraint of this experiment is the limited body shape variation among the actors in the dataset. A larger and more diverse dataset is therefore needed to properly assess the model's shape recovery capabilities. Despite the results in Table 4.2, we conclude that the signals extracted from the 3D keypoints by the TCN do not contain sufficient information to reliably and consistently recover meaningful human body shape in our setup.

4.3 Comparison against VIBE (RQ 3)

How does the performance of a human mesh reconstruction pipeline based on sparse 3D keypoints compare to a direct video-to-mesh baseline (VIBE)?

Here, SMPL is initialized with both predicted pose and shape parameters, and our full pipeline is compared with VIBE.

VIBE outperforms our method in shape-neutral pose reconstruction (Table 4.1), while our method achieves better shape metrics (Table 4.2). However, the latter is inconclusive due to the small dataset and VIBE's more consistent predictions for the same actor (Figure 4.2). Full-pipeline results are shown in Table 4.3, where our method is labeled *Adapted SMPLR + TCN*. VIBE performs better across all metrics, although the gap is smaller for PA-MPJPE than MPJPE, partly due to the frame mismatch described in Section 2.2.5.

Table 4.3: Full-pipeline SMPL pose reconstruction accuracy (mm).

	PA-MPJPE ↓	MPJPE ↓	PVE ↓
VIBE	52.18	82.75	98.54
Adapted SMPLR + TCN	63.81	101.84	120.62

In addition to reconstruction accuracy, we analyze temporal consistency by measuring the acceleration of the regressed joints. The joint acceleration $\mathbf{a}_{t,j}$ is obtained from the standard second temporal difference of the 3D joint positions $\mathbf{p}_{t,j}$, while the average joint acceleration magnitude A_t is computed over all J joints. We formalize these quantities in Equations 4.1 and 4.2 directly from *Vibe's code implementation*[3], while the corresponding quantities are described conceptually in the VIBE paper [9]:

$$\mathbf{a}_{t,j} = \mathbf{p}_{t-1,j} - 2\mathbf{p}_{t,j} + \mathbf{p}_{t+1,j}, \qquad (4.1)$$

$$A_t = \frac{1}{J}\sum_{j=1}^{J} \|\mathbf{a}_{t,j}\|_2. \qquad (4.2)$$

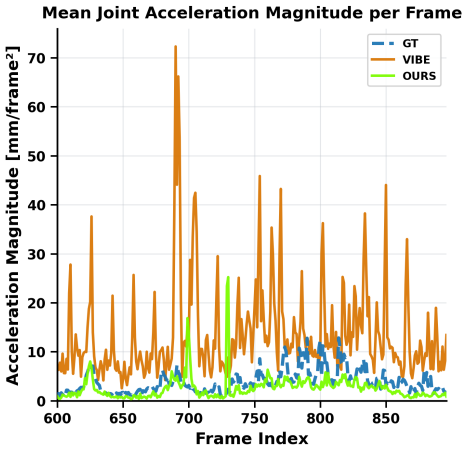


Figure 4.3: Comparison of joint acceleration magnitude between Ground-Truth, VIBE, and our model on sequence *downtown_arguing_oo*.

Figure 4.3 shows the acceleration profile for the representative test sequence *downtown_arguing_00*. VIBE shows significantly higher joint acceleration than the ground-truth, which indicates high-frequency jitter. In contrast, our model is much closer to the ground-truth due to MotionBERT's temporal modeling. However, some frames show oversmoothing, resulting in lower joint acceleration than the ground-truth and potentially lost information.

Figure 4.4 shows two example poses for our model and VIBE. A possible sign of oversmoothing in our method is visible in (a), where the right arm of the person on the left appears more dampened in its upward movement than in VIBE.

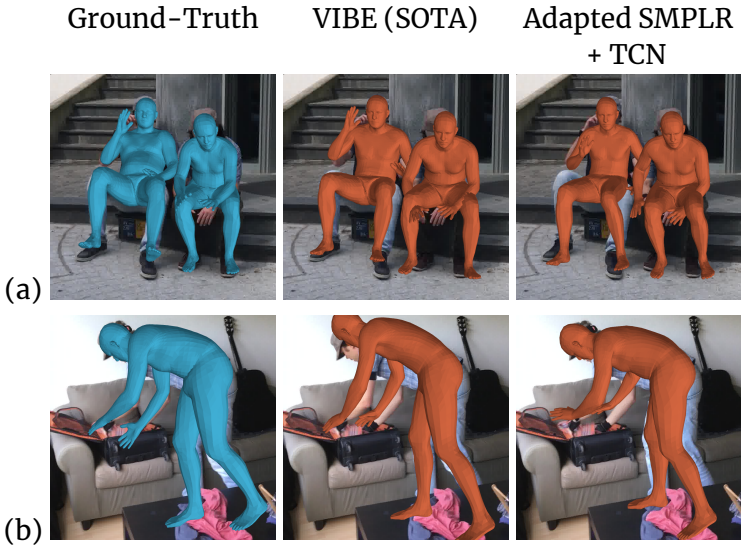


Figure 4.4: Qualitative comparison of Ground-Truth, VIBE (SOTA), and our Deep Learning (proposed) model for two sample frames (predicted pose and shape).

In conclusion, our method achieves better temporal consistency compared to VIBE out of the box, while VIBE achieves better reconstruction accuracy. Preference for one of the two methods is therefore highly dependent on the specific application.

These findings depend on the specific components chosen for our pipeline and our comparison against VIBE. Using alternative models or keypoint estimators may yield different results. Nonetheless, since MotionBERT is a representative, state-of-the-art 3D keypoint estimator, the relative strengths and weaknesses observed here offer a realistic insight into estimating SMPL parameters from sparse keypoints.

Alternative Methods

For the task of human mesh reconstruction, several alternative methods exist. The literature focuses strongly on deep learning approaches, applied directly to image features [2]. As a representative example of this approach, Section 2.2.4 presents VIBE [9].

5.1 HybrIK

When focusing on mesh reconstruction from sparse 3D keypoints, besides deep-learning approaches operating on the 3D skeleton, such as the one proposed in this thesis, inverse-kinematics-based approaches have also been explored for recovering mesh parameters from 3D joint configurations [1].

Inverse kinematics (IK) describe the problem of determining the joint angles of a kinematic chain that achieve a desired configuration of one or more end effectors [29]. In the context of human body reconstruction, IK methods estimate body joint rotations that best match the observed 3D joint positions.

One prominent example which effectively combines deep learning on image inputs with inverse kinematics is Hybrid Analytical-Neural Inverse Kinematics for Body Mesh Recovery (HybrIK) [4]. The authors propose a method that combines analytical inverse kinematics with deep learning to estimate the SMPL parameters that best fit the given 3D keypoints. They decompose body-part rotation into two components: swing and twist. The swing rotation determines the direction of a body part and can be computed directly from the 3D joint locations. The twist rotation describes the rotation around the body part's own axis and cannot be inferred from joint positions alone. HybrIK analytically computes the swing rotations by taking parent-child vectors from predicted 3D human pose coordinates. HybrIK uses Rodrigues' rotation formula [30] to calculate the necessary rotation. This transforms a predefined rest-pose (T-pose) vector configuration, matching the neutrally initialized SMPL model, to the current pose. The twist component is then additionally solved by a neural network, applied directly to extracted image features. The same CNN simultaneously estimates the shape parameters β to complete the SMPL model.

The authors emphasise that by solving the swing rotations analytically and learning only the remaining twist rotations from the 3D keypoints, the regression task is significantly simplified compared to approaches that directly regress SMPL pose parameters from image features [4].

Conclusion and Outlook

6.1 Conclusion

This thesis investigated human body mesh reconstruction from sparse 3D keypoints. The potential of estimating SMPL pose and shape parameters from this limited input data was evaluated by addressing three main research questions:

Research Question 1 (How well can SMPL pose parameters be reconstructed from sparse 3D keypoints using a learned regression network?) explored how well a learned regression network can estimate pose parameters. The results show that the deep learning approach outperformed the optimization-based baseline. Operating on 3D skeletal representations simplifies the regression task compared to learning directly from image features, which allows a simple MLP to learn a plausible mapping from parent-child vectors to SMPL pose parameters. However, the sparse keypoint representation limits reconstruction quality, as finer pose details cannot be recovered. In contrast, the optimization-based baseline was unable to reliably estimate plausible poses, as the underconstrained objective frequently converged to anatomically implausible solutions.

Research Question 2 (Can SMPL body shape parameters be reconstructed from sparse 3D keypoint sequences using a temporal network exploiting motion dynamics?) investigated whether body shape can be inferred from temporal patterns in 3D joint motion. The exploration showed that normalized keypoints contain only weak shape-specific information. Although the proposed temporal CNN produced realistic body shapes, it did not substantially outperform a generic baseline shape. Furthermore, body shape predictions for the same actor varied considerably across different sequences. The low body shape diversity in the 3DPW test dataset also made it difficult to fully assess the model's shape recovery capabilities.

Research Question 3 (How does the performance of a human mesh reconstruction pipeline based on sparse 3D keypoints compare to a direct video-to-mesh baseline (VIBE)?) evaluated how the proposed modular pipeline compares to a direct video-to-mesh baseline. The results show a trade-off between pose accuracy and temporal consistency. While VIBE achieves higher accuracy by a significant margin, our sparse 3D keypoint pipeline produces smoother reconstructions with reduced jitter. However, occasional oversmoothing can be observed, which can lead to a loss of finer pose details.

In summary, this work shows that while traditional approaches based on monocular RGB images still lead in reconstruction accuracy, intermediate sparse 3D keypoints serve as a viable, temporally stable foundation for human mesh reconstruction. Furthermore, the proposed modular pipeline provides flexibility by allowing individual components to be improved or replaced as more advanced methods become available.

6.2 Outlook

Several directions could improve the proposed pipeline. First, using multi-view data could address the depth ambiguity inherent to monocular methods. For a more direct comparison with state-of-the-art approaches, the pipeline could also be extended with a 3D human pose estimation model that predicts keypoints directly in camera coordinates, eliminating the coordinate frame mismatch that currently influences the evaluation. Furthermore, a pose estimation model that preserves consistent relative limb proportions across sequences may provide a stronger cue for body shape estimation than the motion dynamics exploited in this work. Finally, adopting a more advanced pose estimation model that predicts a larger set of keypoints could enable more detailed body reconstruction and facilitate the use of the more elaborate SMPL-X [31] model, extending the representation to additional joints and facial expressions and thereby broadening the range of potential applications.

The training and evaluation could also be extended to include a wider range of datasets with more diverse body shapes, poses, and motion types. This would provide a more comprehensive assessment of the proposed method's generalizability while exposing the models to greater variation during training. Specifically, datasets designed for body shape estimation, as well as datasets containing more challenging and dynamic motions, would be valuable additions. Furthermore, broader evaluations could provide deeper insights into the advantages of mesh reconstruction from sparse 3D keypoints. For example, evaluating computational efficiency and comparing it with existing approaches could help identify additional benefits and limitations.

Although the proposed pipeline does not achieve state-of-the-art accuracy and is unlikely to be sufficient for full human mesh reconstruction in many applications on its own, the use of 3D keypoints as an intermediate representation remains a promising direction. As discussed throughout this thesis, several previous works have demonstrated the benefits of incorporating 3D keypoints alongside image information for human mesh reconstruction. As 3D human pose estimation continues to improve, increasingly accurate joint predictions are likely to further enhance methods that utilize keypoints as an additional source of supervision for human mesh reconstruction.

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